The same friendship network, three community assignments

Spectral
K=11; largest community 91.9%

SCORE
K=11; largest community 18.7%

Louvain
K=11; largest community 23.8%

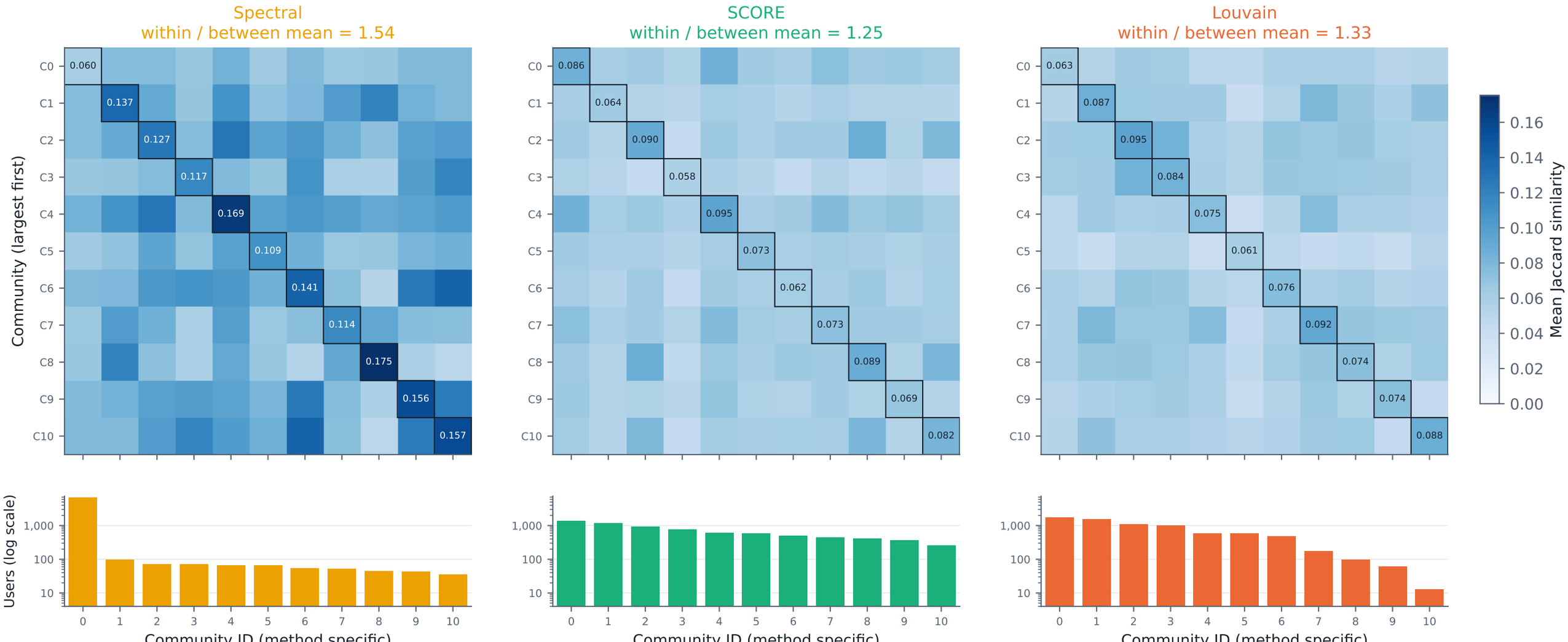


Colors matched to SCORE by maximum membership overlap (visual aid only)

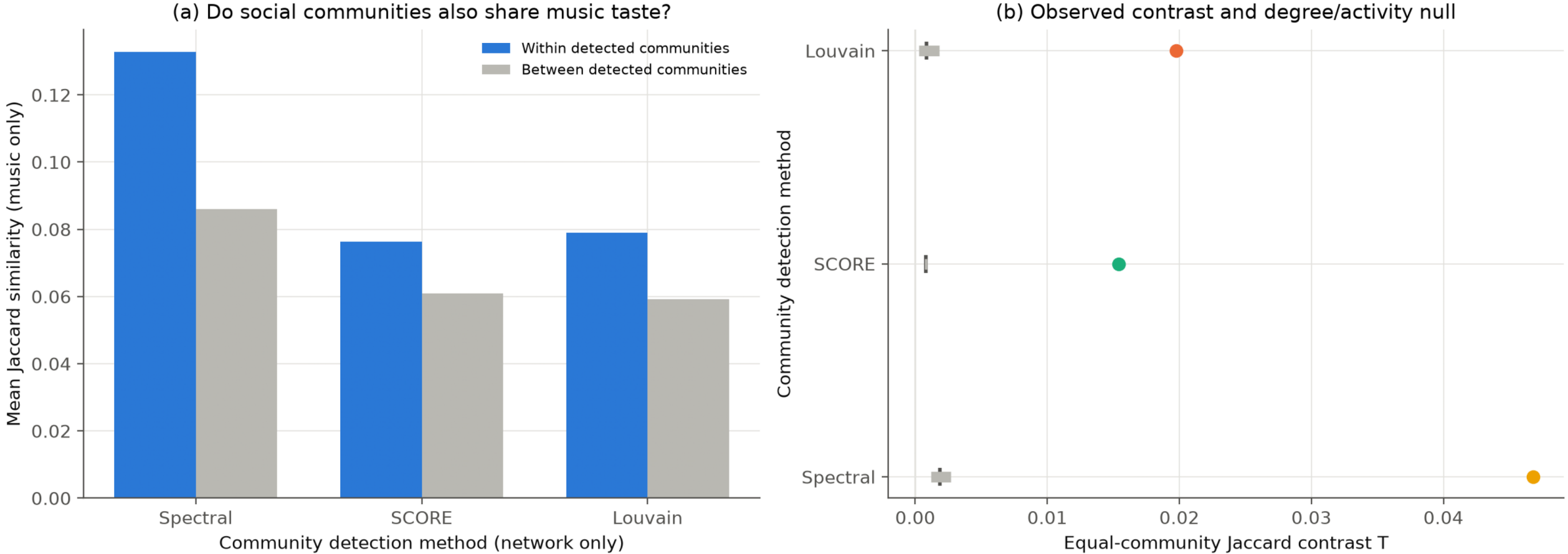
SCORE C0 · SCORE C1 · SCORE C2 · SCORE C3 · SCORE C4 · SCORE C5 · SCORE C6 · SCORE C7 · SCORE C8 · SCORE C9 · SCORE C10

LastFM Asia (March 2020 snapshot): 7,431 users and 27,045 edges. Positions and node sizes are identical. Matching colors does not imply identical communities.

Music similarity inside and between graph-derived communities



LastFM Asia (March 2020 snapshot); same color scale in all panels. Diagonal boxes: within-community means; off-diagonal cells: between-community means.
Community IDs are method specific. Lower panels show sizes; cells are descriptive means, not significance tests.

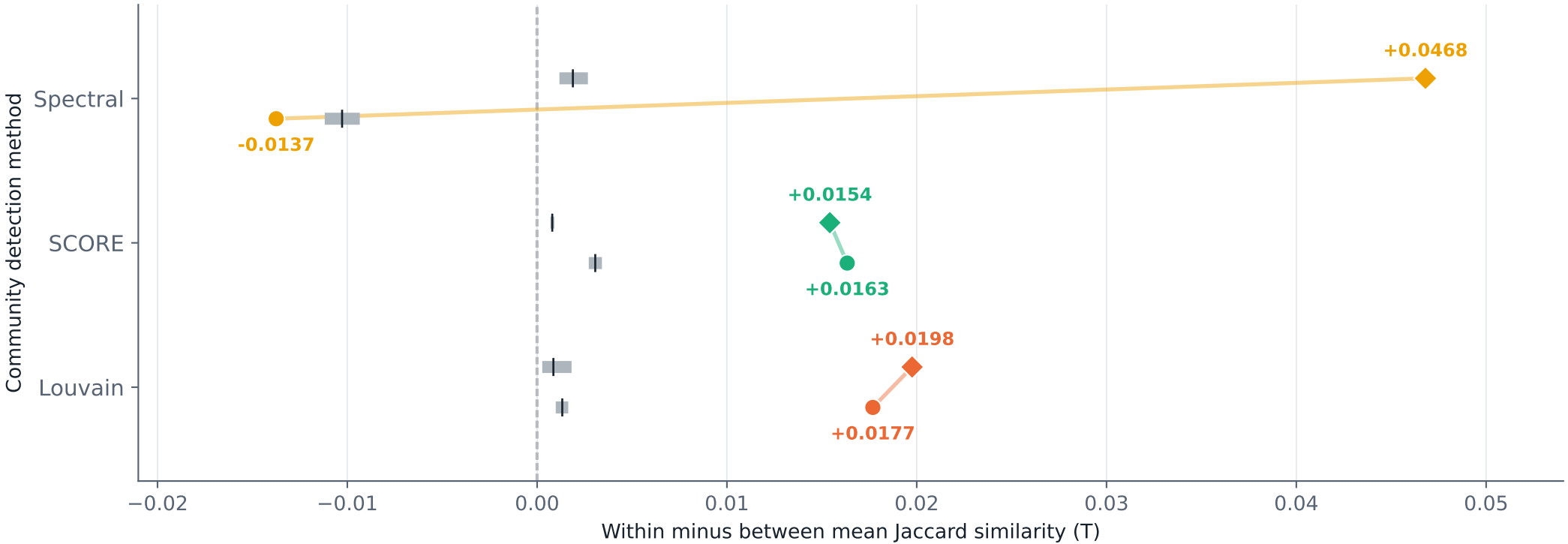


LastFM Asia; B=1999. Points: observed T; grey bars: central 95% permutation ranges, not effect confidence intervals.

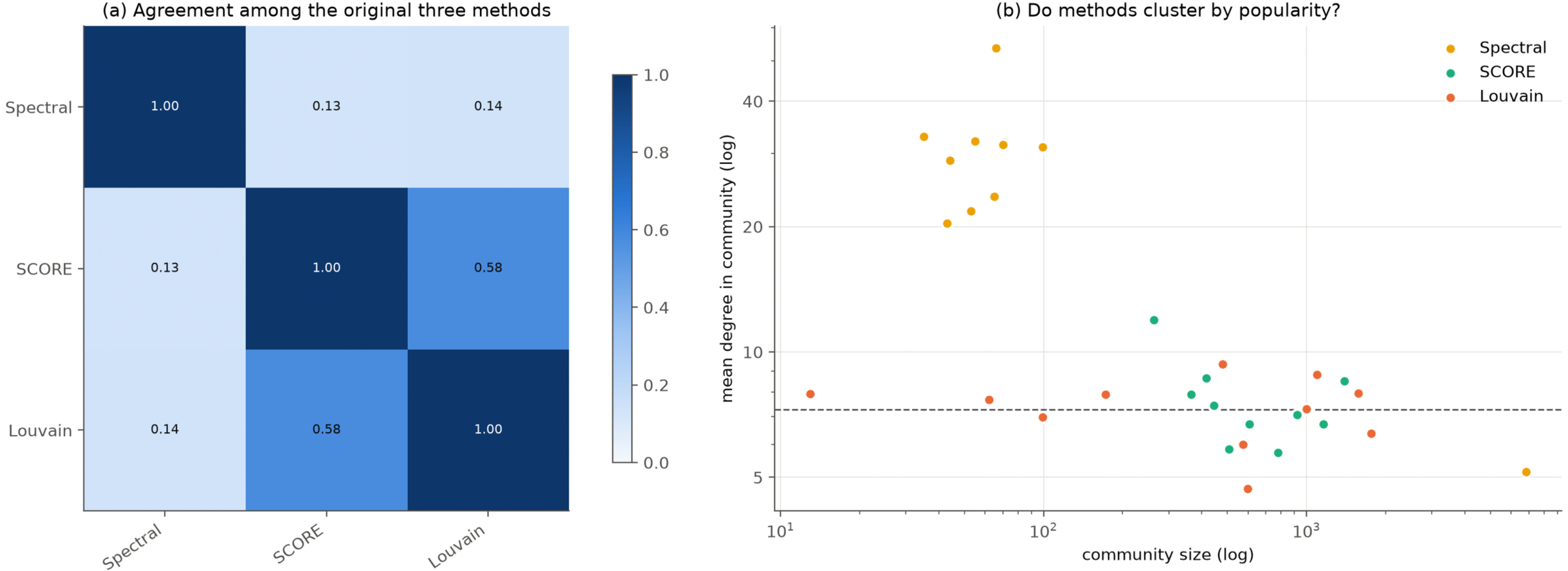
Source: LastFM Asia (March 2020 snapshot). Music evaluates graph-only communities. Permutation ranges describe the conditional null, not uncertainty about a population effect.

Does the conclusion depend on how communities are weighted?

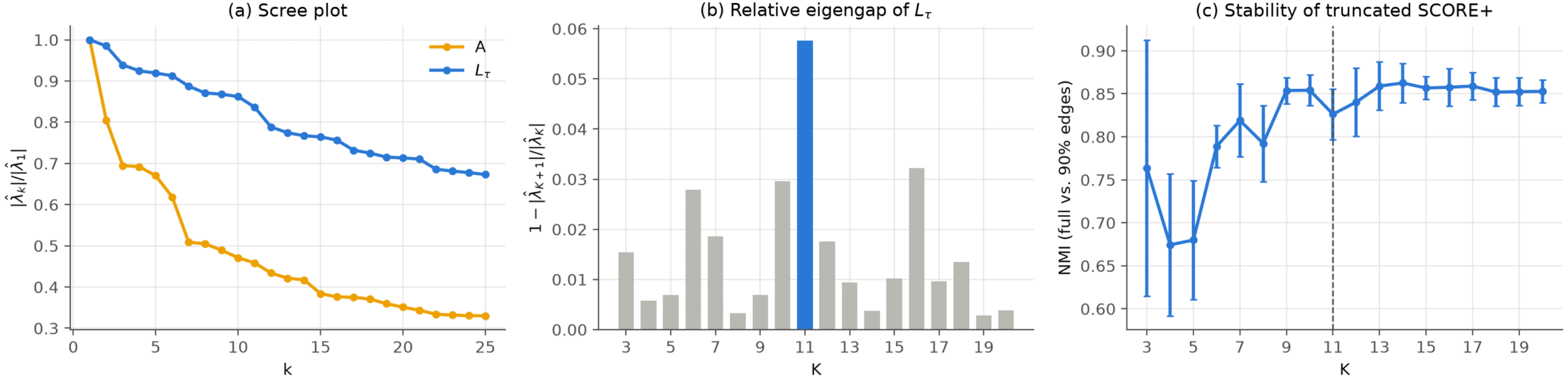
Spectral changes sign; SCORE and Louvain retain positive contrasts under both estimands.



LastFM Asia (March 2020 snapshot); B=1999. Grey ranges are permutation references, not effect confidence intervals.
A negative observed contrast is descriptive; the reported one-sided test targets a positive association.



Source: LastFM Asia (March 2020 snapshot). NMI measures partition agreement, not accuracy. The hub/size comparison is descriptive; real community labels are unknown.



Source: LastFM Asia (March 2020 snapshot). Tuning uses the graph alone. Stability error bars are SD across edge-deletion perturbations, not confidence intervals for true K.